

BMP Image Processing

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

BITMAPFILEHEADER	
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BITMAPINFOHEADER	
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RGBQUAD	
RGB color structure	6

Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

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Chapter 3

Class Documentation

3.1 BITMAPFILEHEADER Struct Reference

BMP file header structure.

```
#include <main.h>
```

Public Attributes

- `uint16_t bfType`
File type (must be "BM")
- `uint32_t bfSize`
File size in bytes.
- `uint16_t bfReserved1`
Reserved (must be 0)
- `uint16_t bfReserved2`
Reserved (must be 0)
- `uint32_t bfOffBits`
Offset to pixel data.

3.1.1 Detailed Description

BMP file header structure.

The documentation for this struct was generated from the following file:

- [main.h](#)

3.2 BITMAPINFOHEADER Struct Reference

BMP information header structure.

```
#include <main.h>
```

Public Attributes

- uint32_t [biSize](#)
Size of this header.
- int32_t [biWidth](#)
Image width in pixels.
- int32_t [biHeight](#)
Image height in pixels (positive = bottom-up)
- uint16_t [biPlanes](#)
Number of color planes (must be 1)
- uint16_t [biBitCount](#)
Bits per pixel (24 for RGB)
- uint32_t [biCompression](#)
Compression type (0 = none)
- uint32_t [biSizeImage](#)
Image size in bytes (0 for uncompressed)
- int32_t [biXPelsPerMeter](#)
Horizontal resolution (pixels/meter)
- int32_t [biYPelsPerMeter](#)
Vertical resolution (pixels/meter)
- uint32_t [biClrUsed](#)
Number of colors in palette.
- uint32_t [biClrImportant](#)
Important colors (0 = all)

3.2.1 Detailed Description

BMP information header structure.

The documentation for this struct was generated from the following file:

- [main.h](#)

3.3 RGBQUAD Struct Reference

RGB color structure.

```
#include <main.h>
```

Public Attributes

- uint8_t [rgbBlue](#)
Blue component (0-255)
- uint8_t [rgbGreen](#)
Green component (0-255)
- uint8_t [rgbRed](#)
Red component (0-255)
- uint8_t [rgbReserved](#)
Reserved (not used in 24-bit BMP)

3.3.1 Detailed Description

RGB color structure.

Note

For 24-bit BMP format (blue, green, red order)

The documentation for this struct was generated from the following file:

- [main.h](#)

Chapter 4

File Documentation

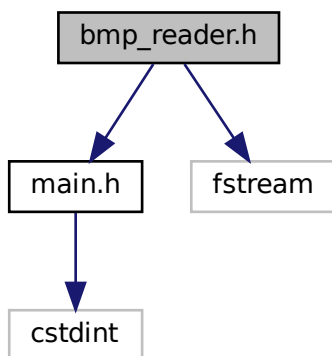
4.1 bmp_reader.h File Reference

BMP file I/O operations.

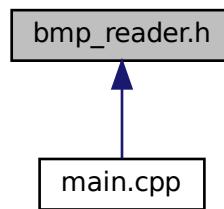
```
#include "main.h"
```

```
#include <fstream>
```

Include dependency graph for bmp_reader.h:



This graph shows which files directly or indirectly include this file:



Functions

- bool `loadBMP` (const char *fileName, `BITMAPFILEHEADER` &fileHeader, `BITMAPINFOHEADER` &fileInfoHeader, `RGBQUAD` **&rgbInfo)
Loads BMP file into memory.
- bool `saveBMP` (const char *fileName, const `BITMAPFILEHEADER` &fileHeader, const `BITMAPINFOHEADER` &fileInfoHeader, `RGBQUAD` **&rgbInfo)
Saves BMP file from memory.
- void `cleanupRGBInfo` (`RGBQUAD` **&rgbInfo, unsigned int height)
Deallocates memory for pixel array.

4.1.1 Detailed Description

BMP file I/O operations.

4.1.2 Function Documentation

4.1.2.1 cleanupRGBInfo()

```
void cleanupRGBInfo (
    RGBQUAD **& rgbInfo,
    unsigned int height )
```

Deallocates memory for pixel array.

Parameters

<i>rgbInfo</i>	Reference to 2D pixel array
<i>height</i>	Image height (number of rows)

4.1.2.2 loadBMP()

```
bool loadBMP (
    const char * fileName,
    BITMAPFILEHEADER & fileHeader,
    BITMAPINFOHEADER & fileInfoHeader,
    RGBQUAD **& rgbInfo )
```

Loads BMP file into memory.

Parameters

<i>fileName</i>	Path to BMP file
<i>fileHeader</i>	Reference to file header structure
<i>fileInfoHeader</i>	Reference to info header structure
<i>rgbInfo</i>	Reference to 2D pixel array

Returns

true if successful, false otherwise

4.1.2.3 saveBMP()

```
bool saveBMP (
    const char * fileName,
    const BITMAPFILEHEADER & fileHeader,
    const BITMAPINFOHEADER & fileInfoHeader,
    RGBQUAD **& rgbInfo )
```

Saves BMP file from memory.

Parameters

<i>fileName</i>	Output file path
<i>fileHeader</i>	File header structure
<i>fileInfoHeader</i>	Info header structure
<i>rgbInfo</i>	2D pixel array

Returns

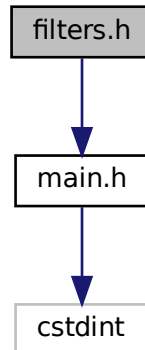
true if successful, false otherwise

4.2 filters.h File Reference

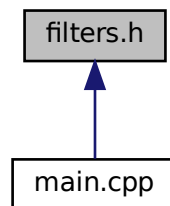
Image filter operations.

```
#include "main.h"
```

Include dependency graph for filters.h:



This graph shows which files directly or indirectly include this file:



Functions

- void [applyGaussianFilter](#) ([RGBQUAD](#) **rgbInfo, unsigned int width, unsigned int height)
Applies Gaussian blur filter to image.

4.2.1 Detailed Description

Image filter operations.

4.2.2 Function Documentation

4.2.2.1 applyGaussianFilter()

```
void applyGaussianFilter (
    RGBQUAD ** rgbInfo,
    unsigned int width,
    unsigned int height )
```

Applies Gaussian blur filter to image.

Parameters

<i>rgbInfo</i>	2D pixel array (input/output image)
<i>width</i>	Image width
<i>height</i>	Image height

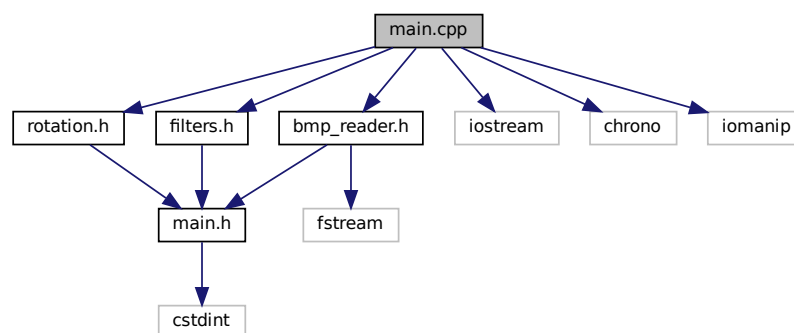
Note

Uses 5x5 kernel with fixed coefficients

4.3 main.cpp File Reference

Main program for image processing.

```
#include "bmp_reader.h"
#include "rotation.h"
#include "filters.h"
#include <iostream>
#include <chrono>
#include <iomanip>
Include dependency graph for main.cpp:
```



Functions

- int **main** (int argc, char *argv[])

4.3.1 Detailed Description

Main program for image processing.

This program processes BMP images by:

1. Rotating 90° clockwise
2. Rotating 90° counter-clockwise
3. Applying Gaussian blur filter

Output files:

- rotated_90_clockwise.bmp
- rotated_90_counterclockwise.bmp
- gaussian_filtered.bmp

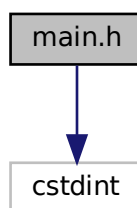
Usage: ./start <input_file.bmp>

4.4 main.h File Reference

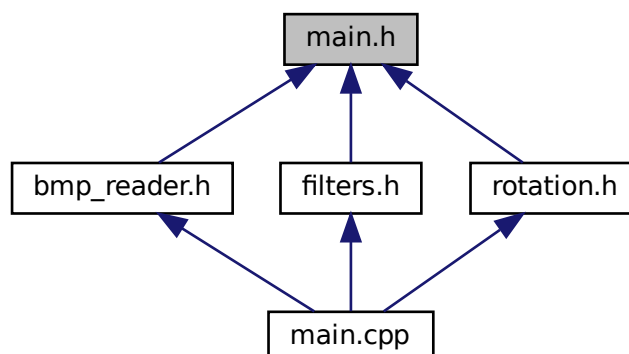
BMP file format structures and definitions.

```
#include <cstdint>
```

Include dependency graph for main.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct [BITMAPFILEHEADER](#)
BMP file header structure.
- struct [BITMAPINFOHEADER](#)
BMP information header structure.
- struct [RGBQUAD](#)
RGB color structure.

4.4.1 Detailed Description

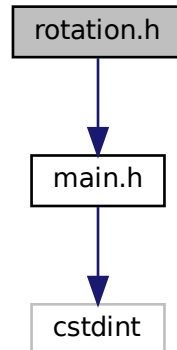
BMP file format structures and definitions.

4.5 rotation.h File Reference

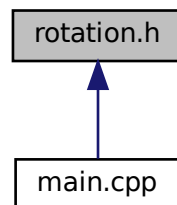
Image rotation operations.

```
#include "main.h"
```

Include dependency graph for rotation.h:



This graph shows which files directly or indirectly include this file:



Functions

- **RGBQUAD ** rotate90Clockwise** (**RGBQUAD ****rgbInfo, unsigned int width, unsigned int height)
Rotates image 90° clockwise.
- **RGBQUAD ** rotate90CounterClockwise** (**RGBQUAD ****rgbInfo, unsigned int width, unsigned int height)
Rotates image 90° counter-clockwise.

4.5.1 Detailed Description

Image rotation operations.

4.5.2 Function Documentation

4.5.2.1 rotate90Clockwise()

```
RGBQUAD** rotate90Clockwise (
    RGBQUAD ** rgbInfo,
    unsigned int width,
    unsigned int height )
```

Rotates image 90° clockwise.

Parameters

<i>rgbInfo</i>	2D pixel array (input image)
<i>width</i>	Image width
<i>height</i>	Image height

Returns

Pointer to new rotated pixel array

4.5.2.2 rotate90CounterClockwise()

```
RGBQUAD** rotate90CounterClockwise (
    RGBQUAD ** rgbInfo,
    unsigned int width,
    unsigned int height )
```

Rotates image 90° counter-clockwise.

Parameters

<i>rgbInfo</i>	2D pixel array (input image)
<i>width</i>	Image width
<i>height</i>	Image height

Returns

Pointer to new rotated pixel array

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